MIL-M-49125(EL)
25 May 1977

MILITARY SPECIFICATION

MOUNTING GROUP, AZIMUTH-ELEVATION, OA-8933( )/GVS-5

This specification is approved for use by the Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the following assemblies of one type of Mounting Group, Azimuth-Elevation, OA-8933( )/GVS-5, used in conjunction with Laser Rangefinder AN/GVS-5( ):

Mount, Azimuth-Elevation, Laser Rangefinder MT-4880( )/GVS-5
Tripod SM-D-882365
Mount, Shelf SM-D-882391
Mounting Bracket, Laser Rangefinder MT-4881( )/GVS-5
Case, Field, Azimuth-Elevation Mount CY-7667( )/GVS-5
Case, Field, Azimuth-Elevation Mount CY-7668( )/GVS-5
Case, Transit Equipment Azimuth-Elevation Mount CY-7669( )/GVS-5 (5.2.6.6)

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-S-781	Strapping, Steel, Flat and Seals
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner

Beneficial comments(recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Command, ATTN: DRSEL-RD-TS-S, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5860

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PPP-B-636	Box, Fiberboard
PPP-F-320	Fiberboard, Corrugated and One Solid, Sheet Stock, Container Grade and Cut Shapes

Military

MIL-P-116	Preservation-Packaging, Methods Of
MIL-P-11268	Parts, Materials and Processes used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-13926	Fire Control Material: General Specification Covering the Manufacture and Inspection of
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-N-49063	Night Vision Sight, Individual Served Weapon, AN/PVS-4
MIL-N-49064	Night Vision Sight, Crew Served Weapon, AN/TVS-5
MIL-P-19644	Plastic Molding Material (Polystyrene Foam Expanded Bead)
MIL-P-19834	Plates, Identification, Metal Foil, Adhesive Backed
MIL-P-26514	Polyurethane Foam, Rigid or Plastic for Packaging

STANDARDS

Federal

FED-STD-356	Commercial Packaging of Supplies and Equipment

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired and other Devices.
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities

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DRAWINGS

US Army Electronics Command

DL-SM-B-852466	Mounting Group, Azimuth-Elevation OA-8933()/GVS-5
DL-SM-B-852482	Mount, Azimuth-Elevation, Laser Rangefinder MT-4880()/GVS-5
DL-SM-B-852568	Mounting Bracket, Laser Rangefinder MT-4881()/GVS-5
DL-SM-B-882396	Tripod
DL-SM-B-882395	Mount, Shelf
DL-SM-B-852484	Case, Field, Azimuth-Elevation Mount CY-7667()/ GVS-5
DL-SM-B-852483	Case, Field, Azimuth-Elevation Mount CY-7668()/ GVS-5
DL-SM-B-852482	Case, Transit, Azimuth-Elevation Mount CY-7669()/GVS-5
*SM-A-882386	
SM-A-882441	

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. Mounting Group, Azimuth-Elevation OA-8933()/GVS-5 is a laser rangefinder azimuth-elevation mount set in a watertight case. Mount, Laser Rangefinder MT-4880()/GVS-5, the principal unit, is used in conjunction with the Laser Rangefinder AN/GVS-5() to provide target elevation and azimuth angle data. An adapter bracket, Mounting Bracket, Laser Rangefinder MT-4881()/GVS-5 is included for operation in conjunction with a night vision device. The mount may also be used in conjunction with a tripod, Tripod SM-D-882365 and a shelf mount, Mount, Shelf SM-D-882391 which are also included in the set.

3.2 First article. When specified in the contract or purchase order, the contractor shall furnish First Article mounts in accordance with 4.3.

3.3 Construction. The mount shall be constructed in accordance with the requirements of this specification and with the drawings listed in USAECOM Data List, DL-SM-B-852466 (including the requirements for parts materials, and processes thereon).

*Classified drawing

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3.3.1 Safety requirements. Safety requirements shall be in accordance with Requirement 1, MIL-STD-454.

3.3.2 Radioactive materials. Radioactive material shall not be used (e.g., luminous dials/markings, electron tubes, surge arrestors and lenses).

3.3.3 Edge rounding. All exposed edges and corners shall be rounded as specified in the equipment drawings.

3.4 Parts, materials and processes; general. In addition to the requirements of this specification the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

3.5 Color and finish. Unless otherwise specified on the applicable drawings, color and finish of metallic parts of the equipment shall be in accordance with MIL-F-14072.

3.6 Marking.

3.6.1 Nameplates. Each nomenclatured item shall be marked with a serialized nameplate conforming to MIL-P-19834.

3.6.2 Marking. The equipment shall be permanently and legibly marked for identification in accordance with MIL-M-13231.

3.7 Weight.

3.7.1 Individual weights. Weights of the individual units shall be as indicated below (See 4.6):

Unit	Weight (maximum)
Mount, Azimuth-Elevation, Laser Rangefinder, MT-4880()/GVS-5	11.60 lbs. (5.26 kg.)
Batteries, BA3030/U (2 each) (see 3.15)	1.1 lbs. (.50 kg.)
Tripod, SM-D-882365	3.0 lbs. (1.36 kg.)
Shelf Mount, SM-D-882391	1.75 lbs. (.79 kg.)
Mounting Bracket, Laser Rangefinder MT-4881()/GVS-5	1.51 lbs. (.68 kg.)
Case, Field Azimuth-Elevation Mount, CY-7667()/GVS-5	8.0 lbs. (3.63 kg.)
Case Field Azimuth-Elevation Mount, CY-7668()/GVS-5	2.25 lbs. (1.02 kg.)
Case, Transit, Azimuth-Elevation Mount, CY-7669()/GVS-5	88.0 lbs. (40.0 kg.)

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3.8 Compatibility.

3.8.1 Universal mounting. The mount shall be physically compatible with the Laser Rangefinder AN/GVS-5. The Laser Rangefinder shall be capable of rigid attachment to and removal from both the Mount MT-4880( )/GVS-5 and the Mounting Bracket MT-4881( )/GVS-5 (See 4.7.2 and 4.7.3)

3.8.2 Night vision sight attachment. The mount shall be physically compatible with Night Vision Sights, AN/PVS-4 (MIL-N-49063) and AN/TVS-5 (MIL-N-49064). These Night Vision Sights shall be capable of rigid attachment to and removal from the Mount, MT-4880( )/GVS-5 (See 4.7.4)

3.8.3 Universal Tripod attachment. The mount shall be physically compatible with the Universal Tripod NSN 6675-00-641-3572. The Mount, MT-4880( )/GVS-5 shall be capable of rigid attachment to and removal from the Universal Tripod by means of a 5/8-11 tapped hole in the center of the mounting surface of the mount (See 4.7.1)

3.8.4 Environment specifications. The equipment shall meet the requirements of 3.8.1, 3.8.2 and 3.8.3 throughout the range of operating environments specified in 3.10.

3.9 Performance Characteristics.

3.9.1 Controls and adjustments. All controls and adjustments shall operate smoothly without binding and with minimum backlash throughout their complete range of movement under all operating conditions.

3.9.1.1 Leveling adjustment screws. The running torque required to rotate each of the three leveling screws shall not exceed 4 inch-pounds when the mount is thermally stabilized at the standard conditions specified in 4.9 and shall not exceed 7 inch-pounds at the operating temperature extremes specified herein.

3.9.1.2 Azimuth indicator rotation (normal operating temperature). The running torque required for rotation of the azimuth indicator shall not be less than 15 inch-pounds, when the indicator latch is in its normal position and the mount is thermally stabilized at the standard conditions specified in 4.9 (See 4.8.1)

3.9.1.3 Azimuth indicator rotation (extreme operating temperature). The running torque required for rotation of the azimuth indicator shall not exceed 12 inch-pounds, when the indicator latch is fully depressed at all operating temperatures specified herein. (See 4.8.1)

3.9.1.4 Azimuth knob rotation. The running torque required to rotate the azimuth adjustment knob to position the azimuth indicator at any position within its travel shall not exceed 32 inch-ounces when thermally stabilized at the standard conditions specified in 4.9 and shall not exceed 50 inch-ounces at the operating temperature extremes specified herein. (See 4.8.1)

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3.9.1.5 Elevation knob rotation. With an actual or simulated worst case load mounted on the elevation gimbal, the running torque required to rotate the elevation adjustment knob to position the elevation indicators at any position within its travel shall not exceed 40 inch-ounces at the operating temperature extremes specified herein. (see 4.8.2)

3.9.2 Accuracy and alignment.

3.9.2.1 Leveling. The vertex of the V-groove mounting for the laser rangefinder shall not deviate more than ±1.1 mils from an initially extablished level when the mount is rotated from its initial azimuth position to any other azimuth position. (See 4.8.4.1)

3.9.2.2 Azimuth accuracy. The accuracy of azimuth reading shall be ±2 mils from any initial setting on the azimuth indicator. (See 4.8.5.1)

3.9.2.3 Laser Rangefinder Mounting interface. With the mount leveled and the elevation indicators at 0 elevation, the laser rangefinder mounting V-groove shall be level within ±2 mils at any fixed azimuth position. (See 4.8.4.2)

3.9.2.4 Elevation accuracy. The accuracy of elevation reading shall be ±3.6 mils with a repeatability of ±2 mils. (See 4.8.5.2)

3.9.2.5 Circular level. The level bubble shall be fully within the inner circle when leveled and shall remain there when the level is rotated through 360 degrees with its mounting surface in contact with a level plane.

3.9.3 Illumination. The azimuth and elevation indicators and the circular level shall be readable to the unaided eye under all conditions from daylight to dark using the internal, continuously adjustable illumination as necessary at all operating temperatures. (See 4.8.3)

3.9.4 Electrical continuity. The conductors shall be electrically continuous (nominally 0 resistance) between appropriate terminations as specified on the schematic drawing. (See 4.8.6.1)

3.9.5 Insulation resistance. The insulation resistance from the conductor to any part of the mount shall not be less than 1000 megohms, when subjected to a minimum potential of 500 volts, direct current. (See 4.8.6.2)

3.9.6 Visual detectability. With the illumination level set at a minimum consistent with good readability by a fully dark adapted operator viewing the azimuth and elevation indicators at a distance of 10 ±2 inches, the mount shall not be visible in a completely dark environment by a fully dark adapted observer having normal 20/20 vision when viewing the mount at a distance of 25 feet and within a quadrant centered along its line of sight. (See 4.8.7)

3.10 Service conditions. The equipment shall be capable of meeting the following environmental conditions:

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3.10.1 Altitude. The equipment shall be operable without degradation in specified performance at altitudes up to 10,000 feet (3000m) above sea level. It shall withstand air transportation at altitudes to 40,000 feet (12,200m) in its transit case without degradation in specified performance (See 4.10.1).

3.10.2 Temperature.

3.10.2.1 Operating. The mounting group shall be operable without degradation in specified performance at ambient air temperatures from -46°C (-50°F) without solar radiation to 52°C (+125°F) plus the effects of solar radiation at a rate of 1135W/m^2 (360 BTU/FT^2-hr) (See 4.10.2).

3.10.2.2 Storage. The mount in its transit case shall experience no damage or degradation in performance after storage at ambient air temperatures from -57°C (-70°F) to 71°C (+160°F) (See 4.10.2).

3.10.3 Humidity. The equipment shall be operable without degradation in specified performance and shall sustain no physical damage during and after prolonged exposure to relative humidities as high as 94± 4% at ambient air temperatures as high as 69°C (+149°F). Fogging on the inside of glass or optical surfaces of instruments shall not occur (See 4.10.4).

3.10.4 Leakage (immersion). Mount, MT-4880( )/GVS-5 and Case, CY-7669( )/GVS-5 shall each be immersion proof to a covering depth of 3 feet (0.91m) of water for a period of not less than 2 hours (see 4.10.6).

3.10.5 Dust. The mounting group shall withstand in both operating and non-operating conditions, exposure to fine dust particles with wind speeds up to 1750 ft./min. (8.9 m/sec) (See 4.10.3).

3.10.6 Salt fog. The mounting group in operating and non-operating condition shall withstand exposure to marine atmospheres and salt fall-out overland areas (See 4.10.5).

3.10.7 Fungus. The mounting group or material samples used in its design and manufacture shall be resistant to fungus growth encountered worldwide (See 4.10.10).

3.10.8 Vibration and shock. The mounting group shall withstand vibration and shock induced during field transport and rough handling encountered in field operations as well as logistical transportation, handling, and servicing (See 4.10.7, 4.10.8 and 4.10.9).

3.11 Nuclear survivability. All contractor furnished mechanical configurations, electronic assemblages, electronic equipment, electronic circuits and electronic components shall withstand the nuclear environment as specified in Drawing SM-A-882386 (confidential). The equipment shall meet the operational requirements of SM-A-882441 immediately after being subjected to the specified nuclear environments (See 4.11).

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3.12 Maintainability. The equipment shall provide for rapidity and ease of maintenance. Limited-life parts shall be mounted to facilitate ease of replacement. Necessary preventative maintenance information (type, procedure, and period) shall be specified.

3.13 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall conform to Requirement 7 of MIL-STD-454 (See 4.12).

3.14 Human factors engineering. Design configuration as affecting operation, maintainability, and handling shall be maintained through compliance of configuration criteria to the requirements of MIL-STD-1472.

3.15 Government furnished property(see 6.2). Government property used with this equipment consists of the following:

Battery (2 per unit)

Battery type BA3030/U. In temperate conditions, Battery BA30 can be used.

3.16 Government loaned property. Unless otherwise specified in the contract or purchase order, the Government will provide the following items for use in the quality assurance tests of Section 4:

NSN	Nomenclature	Quantity
5855-00-629-5329	Night Vision Sight AN/TVS-5	3
5855-00-629-5343	Night Vision Sight AN/PVS-4	3
6135-00-930-0030	Battery BA-3030/U	As req'd
---	Rangefinder Laser AN/GVS-5( )	2
6675-00-641-3572	Universal Tripod	2

3.17 Workmanship. The equipment including all components and parts shall be manufactured and finished to ensure uniform quality and freedom from defects. Workmanship shall conform to all applicable portions of MIL-P-11268 and Requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract of purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Inspections covered by subsidiary documents (See 4.4).
c. Quality conformance inspections

(1) Quality conformance inspection of equipment before packaging (See 4.5).

(2) Quality conformance inspection of packaging (See 4.15).

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor (see 3.2).
4.3.1 First article units. The contractor shall furnish three first article units of the Mounting Group, Azimuth-Elevation OA-8933( )/GVS-5, unless otherwise specified in the contract. (see 6.3.)

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in Table I and shall be performed in the order specified.

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TABLE I. First article inspection

Item	Inspections	Reqt. Para.	Test Para.	Order of Tests 2/ Unit 1	Unit 2	Unit 3
1.	Inspections covered by subsidiary documents	3.3,3.4 3.5,3.6	4.4	Inspections to be performed on all units 1/		
2.	Group A	See Table II		Inspections to be performed on all units 1/		
3.	Group B	See Table III		Inspections to be performed on all units 1/		
4.	Nuclear survivability	3.11	4.11			2
5.	Maintainability	3.12		5		
6.	Environmental					
	Altitude	3.10.1	4.10.1	1		
	Temperature	3.10.2	4.10.2	2		
	Humidity	3.10.3	4.10.4	3		
	Leakage (Immersion)	3.10.4	4.10.6		1,4 4/	
	Dust	3.10.5	4.10.3		6	
	Salt fog	3.10.6	4.10.5		5 3/	1 3/
	Fungus	3.10.7	4.10.10			
	Vibration	3.10.8	4.10.8,4.10.7		2	
	Shock	3.10.8	4.10.9		3	
7.	Visual detectability	3.9.6	4.8.7	4		

Notes:

1/ The inspections 1 through 3, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the Table.

2/ The numbers in the unit number columns in the table for inspection 4 through 7 specify the order of inspections for the indicated unit.

3/ The mount shall be thoroughly washed, cleaned, dried and refurbished after the fungus and the salt spray tests.

4/ The immersion test shall be performed prior to and following vibration and shock testing of the same test item.

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4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the specification required by this specification,and the inspection requirements specified in the contract or purchase order:

Item	Where Required
Construction	3.3
Materials	3.4
Color and finish	3.5
Markings	3.6

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4, and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specifiction requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government, at its discretion may perform all or any part of the specified inspections to verify the contractor's compliance with the requirements. Test equipment shall be made available by the supplier for Government inspection.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to all inspection and test requirements of Table II Discrete lots shall be formed from units that pass this inspection. Lots of conforming units shall be verified by the supplier utilizing the sampling procedures of MIL-STD-105. For Contract verification inspection, unit shall be defined as each nomenclatured item listed in Table II, and contract verification shall be at unit level. Factors of lot composition not defined herein, or in the contract, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105 and using the AQL's indicated in Table II of this specification.

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TABLE II. Group A inspection.

Inspection	Reqt. Para.	Insp. Para.	Insp. Criteria	AQL 3/ Major	AQL 3/ Minor
Visual and Mechanical 1/					
Mount, Azimuth-Elevation, Laser Rangefinder MT-4880( )/GVS-5	3.3 3.9.1	4.8 and 4.13	100%	1	4
Tripod DL-SM-B-882396	3.3	4.13	100%	1	4
Shelf Mount DL-SM-B-882395	3.3	4.13	100%	1	4
Mounting Bracket, Laser Rangefinder MT-4881( )/GVS-5	3.3	4.13	100%	1	4
Case, Field, Azimuth-Elevation Mount CY-7667( )/GVS-5	3.3	4.13	100%	1	4
Case, Field, Azimuth-Elevation Mount CY-7668( )/GVS-5	3.3	4.13	100%	1	4
Case, Transit, Azimuth-Elevation Mount CY-7669( )/GVS-5	3.3	4.13	100%	1	4
Unit Performance 1/					
Accuracy and alignment	3.9.2	4.8.4,4.8.5	100%	1	2/
Illumination	3.9.3	4.8.3	100%	1	2/
Electrical continuity	3.9.4	4.8.6.1	100%	1	2/
Insulation resistance	3.9.5	4.8.6.2	100%	1	2/

1/ The inspections shall be conducted at standard conditions specified in 4.9.

2/ All defects are major.

3/ AQL is percent defective.

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4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government except that operational inspections in 4.8 shall be last.

4.5.2 Group B inspection. Group B inspection shall be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105.

TABLE III. Group B Inspection.

Inspection	Reqt. Para.	Insp. Para.	AQL Percent Defective
Compatibility	3.8	4.7	4.0
Interchangeability (all units)	3.13	4.12	6.5
Weight (all units)	3.7	4.6	4.0

4.5.2.1 Group B Sampling Plans. The Group B sampling plans for the AQLs listed in Table III shall be as follows.

AQL	Inspection Level
6.5%	S-4
4.0%	S-4

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspections shall consist of the inspections specified in Table IV. Samples shall be selected in accordance with 4.5.3.1.

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TABLE IV. Group C inspection.

Inspection	Reqt. Para.	Inspection Para.
Subgroup I		
Dust	3.10.5	4.10.3
Altitude	3.10.1	4.10.1
Temperature	3.10.2	4.10.2
Humidity	3.10.3	4.10.4
Fungus	3.10.7	4.10.10
Subgroup II		
Leakage (Immersion)	3.10.4	4.10.6
Vibration	3.10.8	4.10.8
Shock	3.10.8	4.10.9
Bounce-loose cargo	3.10.8	4.10.7
Salt fog	3.10.6	4.10.5

4.5.3.1 Sampling for Group C inspection. Four (4) samples of the mounting group shall be selected at random, two (2) samples for each subgroup in Table IV, from the first fifty (50) production units and thereafter from every 300 production units or fraction thereof greater than or equal to one-half (½). If the fraction is less than one-half (½), the selection of Group C samples for the preceding test shall be from units which include the remaining fraction.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in the order specified in Table IV.

4.5.3.3 Group C failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspections may be accepted on the contract or purchase order provided all visible and known damage is repaired and the sample units are resubjected to and pass Group A inspection.

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4.6 Weight. The equipment shall be weighed on a calibrated scale and the weights shall not exceed those specified in 3.7

4.7 Compatibility.

4.7.1 Mount to Universal Tripod. The equipment shall meet the requirements specified in 3.8.3.

4.7.2 Laser rangefinder to mount. The equipment shall meet the requirements specified in 3.8.1.

4.7.3 Laser rangefinder to adapter. The equipment shall meet the requirements specified in 3.8.1.

4.7.4 Night vision sights to mount. The equipment shall meet the requirements specified in 3.8.2.

4.7.5 Operational. With the mount attached to either the tripod or shelf mount, attach the adapter mount, the Laser Rangefinder to the adapter mount, and either Night Vision Sight to the mount. All controls and adjustments shall be operable as specified in 3.9.1 and the operating equipment shall be capable of being boresighted to each other.

4.8 Performance Characteristics.

4.8.1 Azimuth operation. The maximum running torque required to rotate the azimuth knob and azimuth inidcator shall be measured by a torque wrench and appropriate adapter, and shall not exceed the values specified in 3.9.1.2, 3.9.1.3, and 3.9.1.4.

4.8.2 Elevation operation. With an actual or simulated worst case load mounted on the elevation gimbal, the maximum running torque required to rotate the elevation knob shall be measured by a torque wrench and appropriate adapter, and shall not exceed the values specified in 3.9.1.5.

4.8.3 Illumination.

4.8.3.1 High ambient. Under an illumination of at least 100 foot-candles, the azimuth and elevation indicators and the circular level shall be readable as specified in 3.9.3.

4.8.3.2 Low ambient. In a completely dark environment, the illumination shall be sufficient to allow reading of the azimuth and elevation indicators and the circular level as specified in 3.9.3.

4.8.3.3 Transitory ambients. The azimuth and elevation indicators and the circular level shall be readable as specified in 3.9.3 for all light ambients between those of 4.8.3.1 and 4.8.3.2.

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4.8.4 Leveling. Perform the following tests using a precision level reading one minute of arc per division. Attach this level parallel to the vertex of the laser rangefinder mounting V-groove using a suitable adapter.

4.8.4.1 Level accuracy. Attain the best level possible using the precision level, the level adjustment screws, and the elevation adjustment knob. The level shall be maintained within the limits specified in 3.9.2.1 at all azimuth positions.

4.8.4.2 Level alignment. Set the elevation indicator to 0 elevation and attain the best level possible using the circular level and the level adjustment screws. The level shall be maintained within the limits specified in 3.9.2.3 at any fixed azimuth position.

4.8.5 Accuracy. Perform the following tests using a precision rotary table reading to at least 5 arc seconds. A mirrored test fixture, whose optical normal is parallel within 10 arc seconds of the vertex of the laser rangefinder V-groove when mounted, is used in combination with an autocollimator set nominally perpendicular to the rotary table axis.

4.8.5.1 Azimuth accuracy. The mount is attached to the rotary table with the azimuth axis nominally parallel to the rotary table axis and the mount leveled as in 4.8.4. Autocollimate off the mirror and set azimuth indicator to 0 by depressing the azimuth indicator latch. Using the azimuth knob, rotate the azimuth indicator to any multiple of 1 mil. Rotate the rotary table to bring mirror into collimation. The difference between the rotary table and azimuth rotation angles shall not exceed that specified in 3.9.2.2.

4.8.5.2 Elevation accuracy. The mount is attached to a right angle bracket on the rotary table with the azimuth axis nominally perpendicular to the rotary table axis and the leveling adjustment as previously set in 4.8.5.1. Set the elevation indicator to 0 and autocollimate off the mirror by rotating the azimuth and rotary table axes. Using the elevation knob, rotate to any multiple of 1 mil. Rotate the rotary table to bring mirror into autocollimation. The difference between the rotary table and azimuth rotation angles shall not exceed that specified in 3.9.2.4.

4.8.6 Electrical.

4.8.6.1 Continuity. Standard measuring equipment shall be used to determine the electrical continuity and the requirements specified in 3.9.4 shall be met.

4.8.6.2 Insulation resistance. A megohm bridge or other appropriate method shall be used to determine the insulation resistance and the requirements specified in 3.9.5 shall be met.

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4.8.7 Visual detectability. The mount shall meet the requirements specified in 3.9.6.

4.9 Standard test conditions. Unless otherwise specified, inspections shall be performed at room temperature of 25° ± 5°C (77 ± 10°F) and humidity not to exceed 90 percent.

4.10 Service test. (Environmental conditions.) Unless otherwise specified, the service conditions test shall be conducted in accordance with MIL-STD-810. Tests and inspections shall be conducted as specified in Table V.

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TABLE V. Verification test schedule.

Verification Tests Specification Paragraph 4.8.1 thru 4.8.5	Altitude P.4.10.1			High Temp P.4.10.2.1			Low Temp. P.4.10.2.2			Dust P.4.10.3			Humidity P.4.10.4			Leakage P.4.10.4			Vibration P.4.10.7 & P.4.10.8			Shock P.4.10.9		
	Pre Test	During	Post Test	Pre Test	During	Post Test	Pre Test	During	Post Test	Pre Test	During	Post Test	Pre Test	During	Post Test	Pre Test	During	Post Test	Pre Test	During	Post Test	Pre Test	During	Post Test
Azimuth Operation	X	X 1/	X	X	X 2/	X 3/	X	X 4/	X 5/	X		X 4/	X 6/	X 6/	X	X		X	X		X	X		X
Elevation Operation	X	X 1/	X	X	X 2/	X 3/	X	X 4/	X 5/	X		X 4/	X 6/	X 6/	X	X		X	X		X	X		X
Illumination				X	X 2/	X 3/	X	X 4/	X 5/	X		X 4/							X		X	X		X
Azimuth Accuracy				X	X 2/	X 3/	X	X 4/	X 5/										X		X	X		X
Elevation Accuracy				X	X 2/	X 3/	X	X 4/	X 5/										X		X	X		X
Level Accuracy and Level Alignment				X	X 2/	X 3/	X	X 4/	X 5/										X		X	X		X

1/ Conducted during Step 2, Proc 1, MIL-STD-810.
2/ Conducted during Step 8, Proc 2, MIL-STD-810.
3/ Conducted during Step 10, Proc 2, MIL-STD-810.
4/ Conducted during Step 5, Proc 1, MIL-STD-810.
5/ Conducted during Step 7, Proc 1, MIL-STD-810.
6/ Conducted during Step 4,6,8, Proc III, MIL-STD-810.

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4.10.1 Altitude test. The altitude test shall be conducted in accordance with Method 500.1 Procedure I of MIL-STD-810. The equipment shall be subjected to a simulated altitude of 40,000 feet above sea level for one hour in its transit case in a non-operating mode. The chamber pressure shall then be increased to simulate an altitude of 10,000 feet and an operational test performed in accordance with Table V.

4.10.2 Temperature.

4.10.2.1 High temperature test. The High Temperature Test shall be conducted in accordance with Method 501.1 Procedure II of MIL-STD-810. The chamber temperature in step 7 will be maintained at 71°C (160°F) until stabilization is reached and an operational test performed in accordance with Table V.

4.10.2.2 Low temperature test. The low temperature test shall be conducted in accordance with Method 502.1 Procedure I, of MIL-STD-810. The unit shall then be subjected in step 2 to a storage temperature of -56°C (-70°F) for a period of 2 hours following test item temperature stabilization. The chamber temperature shall then be increased to -96°C (-50°F) in step 4 and an operational test performed in accordance with Table V.

4.10.3 Dust test. The dust test shall be conducted in accordance with Method 510.1 Procedure I of MIL-STD-810. A visual inspection for mechanical degradation and an operational test in accordance with Table V shall be performed at the conclusion of the test. The mounting group shall be oriented so that unprotected optical surfaces shall face in the opposite direction to the dust stream for the first three hours of Step 1. During the second three hours of Step 1, the test items shall be reoriented so that the optical surfaces face into the dust stream and the velocity of the air shall be reduced to 300 ± 200 ft/min (1.5 ± 1m/sec) for the remainder of Step 1. Prior to conducting Step 2, the test item shall be reoriented to again face away from the dust stream. Step 2 conditions shall be held for 16 hours. The remainder of the dust test shall be as specified in MIL-STD-810.

4.10.4 Humidity Test. The humidity test shall be conducted in accordance with Method 507.1, Procedure II of MIL-STD-810. The unit shall be operated during the last 5 hours of the last cycle in accordance with Table V.

4.10.5 Salt fog test. The salt fog test shall be conducted in accordance with Method 509.1 Procedure I, of MIL-STD-810. A visual inspection shall be performed at the conclusion of the salt fog test.

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r mechanical failure assembly and loudspeaker mount shall show no evidence o

4.10.6 Immersion test. Mount, MT-4880( )/GVS-5 and equipment case CY-7669( )/GVS-5 and any other support accessories required to be waterproof shall be tested in accordance with Method 512.1, Procedure 1, of MIL-STD-810. All test items required to be waterproof may be tested at the same time up to the capacity of the immersion tank, but must be physically separated from each other and conform as a minimum, to the immersion depth given in the test procedure. A visual inspection for any evidence of leakage and an operational test in accordance with Table V shall be performed at the conclusion of this test.

4.10.7 Bounce-loose cargo. The Mount MT-4880( )/GVS-5 and all support accessories in case CY-7669( )/GVS-5 shall be subjected to Method 514.2 Procedure XI, Part 2 of MIL-STD-810. An operational test in accordance with Table V shall be performed prior to and following this test.

4.10.8 Vibration. The mount MT-4880( )/GVS-5 and all support accessories in case CY-7669( )/GVS-5 shall be subjected to the test of Method 514.2, Procedure X. The vibration level shall be in accordance with Figure 514.2-7, Curve AW. The cycling and sweep time shall be in accordance with Table 514.2-VII. An operational test in accordance with Table V shall be performed prior to and following this test.

4.10.9 Shock Test.

4.10.9.1 Transit drop test. The mount MT-4880( )/GVS-5 and support accessories in case CY-7669( )/GVS-5 shall be tested in accordance with Method 516.2, Procedure II of MIL-STD-810. An operational test in accordance with Table V shall be performed prior to and following this test.

4.10.9.2 Bench handling test. The mounting group or portions thereof requiring bench servicing shall be tested in accordance with Method 516.2, Procedure V of MIL-STD-810. An operational test in accordance with Table V shall be performed prior to and following this test.

4.10.10 Fungus. The fungus test shall be conducted in accordance with Method 508.1, Procedure I of MIL-STD-810. A visual inspection shall be performed at the conclusion of this test.

4.10.11 Failure criteria. Any unit which does not successfully pass the verification test schedule of Table V during and upon completion of any environmental test shall be considered a failure.

4.10.11.1 Failure criteria salt fog. Operation of the equipment is not required. Failure criteria shall be limited to corrosion, deterioration, or change in tolerance limits of any internal or external parts which could in any manner prevent the test item from meeting operational service or maintenance requirements.

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4.10.11.2 Failure criteria fungus. The test item shall show only sparse microbial growth with restricted tuberal growth development in an area of 1 to 10 percent or less of the total area and no more than six unrelated minute colonies with mycelial development in areas other than critical circuit areas such as terminal spacing, etc., with sparse growth due to random contamination or traces of unmixed material ingredients. No operational test is required after this test. The test item shall fail if it shows more than the growth specified above.

4.11 Nuclear survivability. Nuclear Survivability Test shall be conducted in accordance with Drawing SM-A-882441.

4.12 Interchangeability. The mechanical dimensions and electrical performance of parts shall be physically and electrically interchangeable among the unit(s). Mechanical dimensions and electrical performance required by the equipment drawings shall be measured to determine compliance with the interchangeability requirements of 3.13.

4.13 Visual and Mechanical. Prior to assembly into the unit(s), assemblies and modules shall be examined for compliance with the equipment drawings. All external moveable controls shall be operated mechanically. Assembled units shall be examined for compliance with equipment drawings, workmanship, and marking requirements. The equipment shall be examined for the defects listed in MIL-STD-252 and MIL-F-13926. An inspection shall be performed to verify compliance with those portions of 3.3.1, 3.3.2, and 3.3.3 which can be determined visually.

4.14 Human engineering verification. Maintenance of design configuration to the requirements of paragraph 3.14 shall be verified by visual and mechanical inspection.

4.15 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine compliance to the requirements of section 5. The rough handling test will be performed only when specified. (See 6.2h).

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or Commercial, as specified (see 6.2f).

5.1.1 Level A.

5.1.1.1 Cleaning. Components of Mounting Group, Azimuth-Elevation, OA-8933( )/GVS-5 shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Components of Mounting Group, Azimuth-Elevation, OA-8933( )/GVS-5 shall be dried in accordance with the applicable procedure of MIL-P-116.

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5.1.1.3 Preservative application. No preservative required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be unit packed method IC-1.

5.1.1.4.2 Mounting Group, Azimuth-Elevation, OA-8933( )/GVS-5. Each mounting group shall be individually unit packed method III as follows: fold Case CY-7668( ) and place in Case CY-7669( ). Strap Tripod Assembly and Mount Shelf Assembly in Case CY-7667( ). Close cover and secure fastener. Strap Mount MT-4880 in Instrument Case and secure cover of case. Position Case CY-7668( ), CY-7667( ), Instrument Case, and Mounting Bracket MT-4881, in recesses provided in Case CY-7669. Close the cover and secure the fasteners. Cushion the transit case on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, designed and fabricated to protect all projections and absorb the shock of impact in handling and transit. As as alternate, cushioning material may be fabricated of plastic material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. Place the cushioned transit case within a close-fitting fiberboard box conforming to PPP-B-636, type CF,class weather-resistant. Place the technical literature,unit packed as specified in 5.1.1.4.1, on top of contents directly under lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B.

5.1.2.1 Cleaning, drying, preservative application and unit packing. Each mount, laser rangefinder shall be cleaned, dried, preserved and unit packed as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be as specified in FED-STD-356.

5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.2f). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Mounting Group, Azimuth-Elevation, OA-8933( )/GVS-5 unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container

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specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.2.2 Level B. A quantity of Mounting Group, Azimuth-Elevation, OA-8933 ( )/GVS-5 unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, domestic type; PPP-B-621, style 4, class 1; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto.

5.2.3 Commercial packing. Packaging shall be in accordance with FED-STD-356.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with FED-STD-356.

6. NOTES

6.1 Intended use. The mount and its subsidiary equipment is used to provide leveling and azimuth and elevation adjustment and indication to a mounted Laser Rangefinder AN/GVS-5( ) alone or in combination with Night Vision Sight, AN/PVS-4 or Night Vision Sight, AN/TVS-5.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.
b. Whether batteries are to be supplied and the quantities involved. Batteries are not furnished by referencing this specification and must be specifically called for in the contract or purchase order.
c. Whether batteries, if supplied, are to be packed separately or in the cases in the quantities of six per equipment.

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d. Whether government loaned property will be supplied and the quantity and schedule for each item. (See 3.16.)

e. That batteries supplied for inspections are not to be delivered.

f. Level A, level B or Commercial preservation and packing (see Section 5).

g. Prior to packing and packaging, each mount shall be examined for absence of batteries.

h. When rough handling test is required. (see 4.15)

i. Marking and shipping requirements.

j. Applicable stock number.

6.3 First article. Procurement documents should specify the following:

a. Number of units, if different (See 4.3.1).
b. Test plan and reports required (See 4.3.3).

6.4 Definitions.

6.4.1 Damage. Breakage, loosening, shifting, evidence of corrosion, or failure of any finish, hardware, connection, or component; leakage or condensation of moisture within the mount; any degradation in performance or ability to function, and any failure of controls or adjustments to perform their function.

6.4.2 Mil. The mil is the standard artillery mil.

6.5 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.5.3.3).

6.6 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design. The supplier should apply for nomenclature in accordance with the applicable clause in the contract or purchase order (see 1.1).

6.7 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

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6.8 Government-furnished and Government-loaned property. The contracting officer should arrange to furnish or loan the property listed in 3.15 and 3.16. (See 6.2)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION ~~MIL-M-49129 MOUNTING GROUP, AZIMUTH-ELEVATION, OA-8933( )/GVS-5~~ (2)

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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